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RESEARCH ARTICLE

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An adaptive sliding mode fault-tolerant control of a quadrotor unmanned aerial vehicle with actuator faults and model uncertainties

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Abstract

An adaptive sliding mode fault-tolerant control strategy is proposed for a quadrotor unmanned aerial vehicle in this article to accommodate actuator faults and model uncertainties. First, a new reaching law is proposed, with which a sliding mode control (SMC) law is constructed. The proposed reaching law is made up of a sliding variable and the distance between it and a designated boundary layer, and it can effectively suppress the unexpected control chattering while preserving the necessary system tracking performance. Then, an adaptive SMC scheme is proposed to further solve the fault and uncertainty compensation problem. The proposed adaptation law helps to prevent overestimation of the adaptive control parameters, as well as avoiding control chattering. Finally, a number of comparative simulation tests are carried out to validate the effectiveness and superiority of the proposed control strategy. The demonstrated quantitative comparison results confirm its advantages.

KEYWORDS

actuator fault, adaptive sliding mode control, fault-tolerant control, model uncertainty, quadrotor UAV

1 | INTRODUCTION

Unmanned aerial vehicles (UAVs) have been increasingly popular and widely employed in the industrial and academic communities in recent years, such as payload transportation,[1,2] autonomous aerial refueling,[3,4] monitoring and surveillance,[5,6] and last-mile delivery.[7,8] Quadrotor UAVs have attracted a lot of attention among the various types of UAVs, due to its ability to perform vertical takeoff and landing and low cost. In addition, many of these applications are deployed in areas with high safety requirements. As a result, both academia and industry are paying more attention to the safety and reliability of quadrotor UAVs.[9-12] One way to retain high safety and reliability of quadrotor UAVs is to design fault-tolerant control (FTC) systems.[13] The increasing demands on high system performance, safety, and reliability have driven active research on advanced FTC approaches for safety-critical systems.[14-18]

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An overview of the existing FTC strategies for quadrotor UAVs is presented in Reference 19. In general, the commonly used FTC strategies for quadrotor UAVs can be divided into two categories: passive and active FTC approaches. In Reference 20, a sliding mode-based passive FTC scheme is designed for a quadrotor UAV to tolerate actuator faults and model uncertainties. In Reference 21, both passive and active FTC methods are designed for a commercial quadrotor under actuator faults based on sliding mode technique, and their advantages and disadvantages are investigated and compared through analytical results. Essentially, passive FTC approaches use the inherent robustness of the designed control algorithm to mitigate prescribed faults and they may sacrifice tracking control performance under fault-free conditions. In contrast to passive FTC approaches, active FTC methods can reconfigure the controller in real time to compensate for faults based on the fault information provided by the fault diagnosis unit. In Reference 22, by combining a recurrent neural network-based fault estimation technique with a sliding mode feedback controller, an active FTC mechanism is created for actuator fault accommodation. In Reference 23, to compensate for the loss of effectiveness actuator defect caused by battery depletion, an active FTC approach based on linear parameter varying control technique is provided. Although active FTC system can effectively compensate actuator faults, its performance heavily depends on the level of accuracy of the estimated fault magnitude. The fault estimation error and delay may degrade the FTC performance. To resolve this issue, many researchers focus on investigating adaptive FTC strategies in recent years. In Reference 24, to compensate the actuator faults for a quadrotor UAV without the necessity for a fault estimation mechanism, a nonlinear adaptive FTC approach is developed for altitude and attitude tracking. In Reference 25, the position and attitude tracking control of a quadrotor UAV is built using an indirect neural network-based adaptive FTC method that can compensate actuator faults and external disturbances. In Reference 26, an adaptive radial basis function neural network fuzzy sliding mode control (SMC) scheme is built to accommodate actuator faults for a coaxial octorotor UAV. In Reference 27, an adaptive fuzzy backstepping FTC scheme is presented for a modified quadrotor UAV to achieve accommodation of actuator faults. Despite the fact that the aforementioned control strategies are capable of achieving desired tracking performance of quadrotor UAVs in the face of actuator faults, the issues of model uncertainties are not adequately investigated.

In addition, fault-tolerance capability is not the only factor to consider when it comes to ensuring the safety control of quadrotor UAVs. When UAVs are deployed for practical applications, many uncertain factors often occur which may affect their safety and reliability. One of the challenges for safety control of quadrotor UAVs is to deal with model uncertainties. To enhance the quadrotor UAVs' capability for tolerating certain level of model uncertainties, a number of control strategies have been developed. In Reference 28, to manage the attitude and position of a quadrotor UAV while attenuating parameter changes and uncertainties, a model-free-based terminal SMC technique is designed. In Reference 29, an adaptive fuzzy terminal SMC strategy is presented to overcome the quadrotor's model uncertainties and provide a robust path tracking performance. In Reference 30, a complementary control strategy based on approximate dynamic programming is developed to improve the quadrotor UAV tracking performance under model uncertainties. It is worth noting that the robust tracking control performance of the above mentioned control strategies can only be achieved under fault-free conditions. To further ensure robustness of the developed control strategies, the uncertain factors including actuator faults and model uncertainties need to be jointly considered.

In this sense, the developed control system for quadrotor UAVs is anticipated to tolerate the adverse impact produced by actuator faults and model uncertainties for the sake of safety control of quadrotor UAVs. In Reference 31, to mitigate actuator faults and model uncertainties for a quadrotor UAV, a dual adaptive FTC strategy is developed. In Reference 32, an active FTC scheme is presented for a quadrotor UAV, in which the model uncertainty is compensated with adaptive neural network mechanism, and actuator fault information is provided by a fault estimation unit and compensated in an active way. In Reference 33, a sliding mode observer is designed to estimate the uncertainties induced by actuator faults and unknown parameters and integrated with the feedback SMC to compensate the adverse effect of actuator faults. SMC is well known for its robustness against the so-called matched uncertainties. It has been successfully applied in a variety of fields.[34,35] However, the robustness of conventional SMC can only be ensured by selecting a large discontinuous control gain, which may lead to unanticipated control chattering.[36] To solve this problem, a number of adaptive SMC approaches for accommodating actuator faults and model uncertainties have been presented. In Reference 37, an adaptive SMC with projection method is designed that ensures constrained estimations of uncertain inertial parameters for a quadrotor. In Reference 38, for a class of nonlinear systems with actuator faults a multivariable SMC scheme with the aid of an adaptive single-hidden-layer feedforward network is designed. As reported in Reference 39, a fractional order SMC with adaptive fuzzy approximation technique is designed for the attitude system of a spacecraft subject to inertia uncertainty and multiple type actuator faults. In Reference 40, to compensate actuator faults and uncertainties for high-speed trains, an adaptive fault-tolerant SMC scheme is presented. In Reference 41, a continuous adaptive integral SMC strategy is built for a space manipulator to achieve trajectory tracking control against actuator uncertainties.

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Although substantial research studies have been done towards safety control of quadrotor UAVs, there still exist several problems that need to be solved in order to enhance the safety and reliability of quadrotor UAVs: (1) most of the previous works using SMC to deal with actuator faults and model uncertainties usually employ a constant discontinuous control gain, which may lead to a conservative tracking performance in order to avoid control chattering; (2) when utilizing adaptive SMC technique to mitigate actuator faults and model uncertainties, the discontinuous control component of SMC is usually overused resulting in unanticipated control chattering; (3) because of the inherent tracking errors, if sliding variable is used for constructing the adaptive schemes to mitigate actuator faults and uncertainties, it may not be able to guarantee the convergence of adaptive control parameters.

In an effort to overcome the aforementioned challenges in achieving safety control of quadrotor UAVs, this article proposes a unique adaptive sliding mode FTC strategy with a new reaching law to jointly compensate both actuator faults and model uncertainties. The major contributions of this article can be summarized as follows:

(1) A new reaching law is proposed in this article, which can adapt to the variations of the distance between the sliding variable and the designated boundary layer. It features that when the sliding variable is far away from the desired sliding surface, a large discontinuous control gain is generated for a faster reaching time, and when the sliding variable approaches to the desired sliding surface, the discontinuous control gain will gradually decrease to zero to suppress control chattering.
(2) In the existing adaptive sliding mode FTC designs[37,42] for quadrotor UAVs, the adaptation law is commonly constructed with sliding variable. Because of the inevitable tracking errors, it is possible that the adaptive control parameters will not be able to converge. To address this issue, inspired by Xu et al.,[43] this article proposes an adaptation law that can prevent overestimation of adaptive control parameters and suppress control chattering.
(3) To achieve actuator fault tolerance, instead of merely adaptively changing the discontinuous control component of SMC in contrast to the works in References 38 and 44, with the developed control strategy, the continuous and discontinuous control components are both constructed with adaptive parameters. In this case, the utilization of the discontinuous control component will be decreased, which can help to prevent control chattering.

The rest of this article is organized as follows. In Section 2, the modeling and system formulation of the researched quadrotor UAV are presented. In Section 3, the detailed design procedures of the proposed adaptive FTC strategy are explained. Then, the effectiveness of the proposed control strategy is validated and compared in Section 4 through a series of simulation tests. Finally, Section 5 summarizes the general conclusions of this article.

2 | PROBLEM FORMULATION

This section describes the mathematical modeling of the quadrotor UAV under consideration, as well as the corresponding system formulation. Figure 1 depicts the researched quadrotor UAV, that is produced by Quanser. It is assembled with four 10-inch propellers driven by brushless motors in a plus configuration. The motor #1 and #2 spin clockwise (CW), while the motor #3 and #4 rotate counter-clockwise (CCW). The motors are connected to electric speed controllers that receive commands from the flight controller in the form of pulse-width modulation (PWM) outputs. The generated thrust of each individual propeller is varied to control the quadrotor UAV.

2.1 | Modeling of quadrotor UAV

Two reference frames, the body-fixed reference frame and the earth-fixed reference frame, are first defined for the purpose to determine the equations of motion of the quadrotor UAV. The center of gravity of the quadrotor UAV is chosen as the origin of the body-fixed reference frame, whereas the takeoff point is chosen as the origin of the earth-fixed reference frame. The axes of the body-fixed reference frame and the earth-fixed reference frame are represented by (o_b, x_b, y_b, z_b) and (o_e, x_e, y_e, z_e), respectively. Concentrating on the fault-tolerant attitude control of the quadrotor UAV, the dynamic equations of motion of the quadrotor UAV are derived as follows using the Newton–Euler formulation:

$$\tau^B = I\dot{\omega}^B + \omega^B \times I\omega^B, \tag{1}$$

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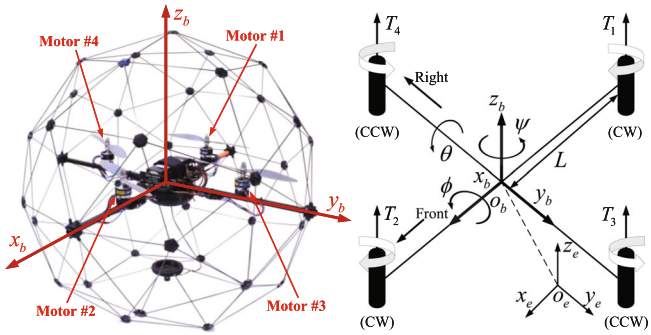

FIGURE 1 Illustration of the researched quadrotor UAV.

where I is a diagonal matrix denoting the quadrotor UAV's moment of inertia that is defined as $I = \mathrm{diag}([I_{xx}, I_{yy}, I_{zz}])$. $\tau^B = [\tau_x, \tau_y, \tau_z]^T$, and $\omega^B = [p, q, r]^T$ represent the vector of resultant torques and the vector of angular velocities, respectively, with respect to the body-fixed reference frame.

The resultant torques operating on the quadrotor UAV are composed of the propeller-generated torques (τ_T) and the rotational motion induced torques (τ_f), that can be formulated as follows:

$$\tau^B = \tau_T + \tau_f = \begin{bmatrix} \tau_\phi \\ \tau_\theta \\ \tau_\psi \end{bmatrix} + \begin{bmatrix} -k_{d1}\dot{\phi} \\ -k_{d2}\dot{\theta} \\ -k_{d3}\dot{\psi} \end{bmatrix}, \tag{2}$$

where $\tau_\phi, \tau_\theta, \tau_\psi$ are the propeller-generated torques along x_b-, y_b-, and z_b-axis, respectively. ϕ, θ, ψ are the roll, pitch, and yaw angle of the quadrotor UAV, and k_{di} $(i = 1, 2, 3)$ is the aerodynamic drag coefficient caused by rotational motion.

The relationship between the propeller-generated torques and the PWM inputs of the electric motors can be formulated as per the illustrated configuration of the quadrotor UAV in Figure 1 as follows:

$$\begin{bmatrix} \tau_\phi \\ \tau_\theta \\ \tau_\psi \end{bmatrix} = \begin{bmatrix} 0 & 0 & K_u L & -K_u L \\ K_u L & -K_u L & 0 & 0 \\ K_y & K_y & -K_y & -K_y \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix}, \tag{3}$$

where u_i $(i = 1, 2, 3, 4)$ is the PWM input of the ith motor whose range is $[0, 0.05]$. A command of 0 corresponds to zero throttle, which will cause the motors to stop. A command of 0.05 corresponds to full throttle. K_u is the thrust gain related to the propeller-generated force, K_y is the torque gain related to the reaction torque generated by the propellers, L is the distance between the individual motor and center of gravity of the quadrotor UAV.

In this sense, substituting Equations (2) and (3) into Equation (1), one can obtain:

$$\begin{cases} \dot{p} = \dfrac{(I_{yy} - I_{zz})qr}{I_{xx}} + \dfrac{K_u L(u_3 - u_4)}{I_{xx}} - \dfrac{k_{d1}\dot{\phi}}{I_{xx}} \\ \dot{q} = \dfrac{(I_{zz} - I_{xx})pr}{I_{yy}} + \dfrac{K_u L(u_1 - u_2)}{I_{yy}} - \dfrac{k_{d2}\dot{\theta}}{I_{yy}} \\ \dot{r} = \dfrac{(I_{xx} - I_{yy})pq}{I_{zz}} + \dfrac{K_y(u_1 + u_2 - u_3 - u_4)}{I_{zz}} - \dfrac{k_{d3}\dot{\psi}}{I_{zz}}, \end{cases} \tag{4}$$

Furthermore, the following equations can be used to define the relationship between the angular velocities and the Euler-angle rates:

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & \sin\phi\tan\theta & \cos\phi\tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi/\cos\theta & \cos\phi/\cos\theta \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix} = R_B^E \begin{bmatrix} p \\ q \\ r \end{bmatrix}, \tag{5}$$

where R_B^E is a transformation matrix from the body-fixed reference frame to the earth-fixed reference frame.

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Assume that the changes of the roll and pitch angles are small to facilitate the controller design. Under this assumption, the transformation matrix R_B^E will be close to an identity matrix, and the angular velocities (p, q, r) can be simply replaced by the Euler-angle rates $(\dot{\phi}, \dot{\theta}, \dot{\psi})$. Note that, this assumption is only made for controller design. In this situation, Equation (4) can be rewritten as follows:

$$\begin{cases} \ddot{\phi} = \dfrac{(I_{yy} - I_{zz})\dot{\theta}\dot{\psi}}{I_{xx}} + \dfrac{K_u L(u_3 - u_4)}{I_{xx}} - \dfrac{k_{d1}\dot{\phi}}{I_{xx}} \\ \ddot{\theta} = \dfrac{(I_{zz} - I_{xx})\dot{\phi}\dot{\psi}}{I_{yy}} + \dfrac{K_u L(u_1 - u_2)}{I_{yy}} - \dfrac{k_{d2}\dot{\theta}}{I_{yy}} \\ \ddot{\psi} = \dfrac{(I_{xx} - I_{yy})\dot{\phi}\dot{\theta}}{I_{zz}} + \dfrac{K_y(u_1 + u_2 - u_3 - u_4)}{I_{zz}} - \dfrac{k_{d3}\dot{\psi}}{I_{zz}}. \end{cases} \tag{6}$$

2.2 | System formulation

Consider a nonlinear quadrotor UAV system with actuator faults and model uncertainties as:

$$\dot{x}(t) = f(x(t), t) + g(x(t), t)B_u L_c(t)u(t), \tag{7}$$

where $x(t) \in \mathbb{R}^n$ is the state vector, $u(t) \in \mathbb{R}^m$ is the control input vector. $f(x(t), t) \in \mathbb{R}^n$ and $g(x(t), t) \in \mathbb{R}^{n\times n}$ represent nonlinear functions with model uncertainties. $B_u \in \mathbb{R}^{n\times m}$ is the control efficiency matrix, which is made up of control input coefficients. $L_c(t) = \mathrm{diag}([l_{c1}(t), l_{c2}(t), \ldots, l_{cm}(t)])$ is a diagonal matrix that represents the level of remaining control effectiveness of the actuators, where $l_{ci}(t)$ $(i = 1, 2, \ldots, m)$ is a scalar satisfying $0 \le l_{ci}(t) \le 1$. If the ith actuator encounters certain level of fault, then $0 \le l_{ci}(t) < 1$, otherwise, the ith actuator works well. In the subsequent sections, the notation (t) is removed for convenience of expression. For example, $u(t)$ is expressed as u.

In order to facilitate the fault-tolerant attitude control design for the quadrotor UAV, define the state vector of the system as $x = [\phi, \dot{\phi}, \theta, \dot{\theta}, \psi, \dot{\psi}]^T$. Thus, the quadrotor UAV system described in Equation (6) can be expressed as a general integral-chain nonlinear system as follows using this state vector:

$$\begin{cases} \dot{x}_{1i} = x_{2i} \\ \dot{x}_{2i} = (h_{1i} + \Delta h_{1i})f_i(x) + (h_{2i} + \Delta h_{2i})x_{2i} + g_i v_i \\ v_i = B_{ui} L_c u, \end{cases} \tag{8}$$

where $i = 1, 2, 3$ represents the roll, pitch, and yaw subsystem, respectively, $u = [u_1, u_2, u_3, u_4]^T$, $x_{1i} = [\phi, \theta, \psi]^T$, $x_{2i} = [\dot{\phi}, \dot{\theta}, \dot{\psi}]^T$. Δh_{1i} and Δh_{2i} represent the model uncertainties of the system induced by the uncertain moments of inertia of the quadrotor UAV, which are bounded as $|\Delta h_{1i}| \le \Delta_1$ and $|\Delta h_{2i}| \le \Delta_2$. For the roll subsystem, $f_1(x) = \dot{\theta}\dot{\psi}$, $h_{11} = (I_{yy} - I_{zz})/I_{xx}$, $h_{21} = -k_{d1}/I_{xx}$, $g_1 = 1/I_{xx}$, $B_{u1} = [0, 0, K_u L, -K_u L]$. For the pitch subsystem, $f_2(x) = \dot{\phi}\dot{\psi}$, $h_{12} = (I_{zz} - I_{xx})/I_{yy}$, $h_{22} = -k_{d2}/I_{yy}$, $g_2 = 1/I_{yy}$, $B_{u2} = [K_u L, -K_u L, 0, 0]$. For the yaw subsystem, $f_3(x) = \dot{\phi}\dot{\theta}$, $h_{13} = (I_{xx} - I_{yy})/I_{zz}$, $h_{23} = -k_{d3}/I_{zz}$, $g_3 = 1/I_{zz}$, $B_{u3} = [K_y, K_y, -K_y, -K_y]$.

Remark 1. The diagonal matrix L_c is mainly used to model the investigated actuator faults. In the context of adaptive FTC framework, the main priority is to compensate the fault and may not need to know the actual faulty level of the actuators. Therefore, the unknown matrix L_c is not used in the developed control law and the following equation $v_i = B_{ui}u$ is employed to calculate the control law.

3 | ADAPTIVE SLIDING MODE FTC STRATEGY

In this section, an adaptive sliding mode FTC strategy is designed to simultaneously compensate actuator faults and model uncertainties for the investigated quadrotor UAV. Figure 2 depicts an overview of the proposed control strategy. To achieve the desired attitude control performance, two issues need to be addressed. The first issue is to design a reaching law-based SMC scheme that ensures the desired system tracking performance while minimizing control chattering under

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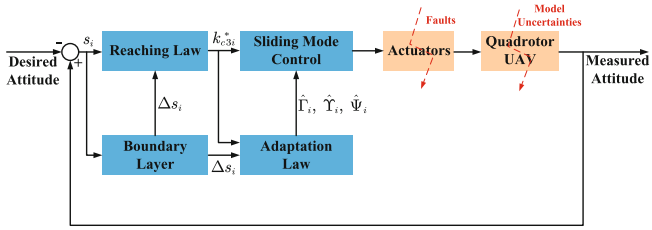

FIGURE 2 Overview of the proposed adaptive sliding mode FTC scheme.

nominal conditions. The second task is to investigate the adaptive control schemes in order to determine the adaptive control parameters without overestimation and then to combine them with the developed control law for accommodation of actuator faults and model uncertainties.

3.1 | Design of reaching law-based SMC

In general, the SMC design typically consists of two steps. The initial step involves establishing a sliding surface on which the system performance can be attained as expected. The second step is to construct an appropriate control law that will push the sliding variable towards the designed sliding surface, ensuring that the system will satisfy the sliding mode reaching requirement.

The desired time-varying system trajectory is denoted as x_{1i}^d, and its derivative is assumed to be bounded. The corresponding tracking error states can be described as $x_{1i}^e = x_{1i} - x_{1i}^d$ and $x_{2i}^e = x_{2i} - \dot{x}_{1i}^d$. With the defined tracking error states, the integral sliding surface for the considered attitude control system of the quadrotor UAV is designed as:[45]

$$s_i = s_{i0} + z_i, \tag{9}$$

where s_{i0} is the linear combination of the tracking error states which is formulated as $s_{i0} = c_i x_{1i}^e + x_{2i}^e$, and z_i includes the integral term that is determined by Equation (10).

$$\begin{aligned} \dot{z}_i &= -c_i x_{2i}^e + k_{c2i} x_{2i}^e + k_{c1i} x_{1i}^e \\ z_i(0) &= -c_i x_{1i}^e(t_0) - x_{2i}^e(t_0), \end{aligned} \tag{10}$$

where k_{c1i} and k_{c2i} denote the adjustable design parameters, and t_0 represents the initial time instant.

Then, combining the above equations, the following equation can be obtained as:

$$\begin{aligned} s_i &= s_{i0} + z_i \\ &= c_i x_{1i}^e + x_{2i}^e - c_i x_{1i}^e(t_0) - x_{2i}^e(t_0) + \int_{t_0}^{t} (-c_i x_{2i}^e(\tau) + k_{c2i} x_{2i}^e(\tau)) d\tau + \int_{t_0}^{t} x_{1i}^e(\tau) d\tau \\ &= c_i x_{1i}^e + x_{2i}^e - c_i x_{1i}^e(t_0) - x_{2i}^e(t_0) - c_i x_{1i}^e + c_i x_{1i}^e(t_0) + k_{c2i} x_{1i}^e - k_{c2i} x_{1i}^e(t_0) + \int_{t_0}^{t} x_{1i}^e(\tau) d\tau \\ &= x_{2i}^e + k_{c2i} x_{1i}^e + k_{c1i} \int_{t_0}^{t} x_{1i}^e(\tau) d\tau - k_{c2i} x_{1i}^e(t_0) - x_{2i}^e(t_0). \end{aligned} \tag{11}$$

Following the construction of the sliding surface, the next step is to design a suitable control law for satisfying the reaching requirement of the sliding mode. The conventional SMC law is usually made up of two control components, namely the continuous control component and the discontinuous robust control component. However, a large discontinuous control gain will lead to the occurrence of control chattering even though it can increase the robustness of the SMC.

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In order to address this problem, this article proposes a novel reaching law as follows:

$$\dot{s}_i = -\frac{k_{c3i}}{\eta_i + (1 + 1/|s_i| - \eta_i)e^{-\alpha_i|\Delta s_i|}}\text{sign}(s_i), \tag{12}$$

where k_{c3i} and α_i are positive design parameters, $0 < \eta < 1$, $\Delta s_i = s_i - \Phi_i \text{sat}(s_i)$, and

$$\text{sat}(s_i) = \begin{cases} \text{sign}(s_i) & \text{if } |s_i| > \Phi_i \\ s_i/\Phi_i & \text{if } |s_i| \leq \Phi_i \end{cases} \tag{13}$$

with Φ_i representing the defined boundary layer thickness.

Remark 2. As can be observed from the afore-defined variable Δs_i, it represents the algebraic distance between the current sliding variable and the designated boundary layer. Δs_i is equal to zero when the sliding variable is inside the designated boundary layer.

Remark 3. With the proposed reaching law, it can be observed from Equation (12) that if Δs_i increases, the function $k_{c3i}/[\eta_i + (1 + 1/|s_i| - \eta_i)e^{-\alpha_i|\Delta s_i|}]$ will converge to the value of k_{c3i}/η_i, which is larger than k_{c3i}. As can be seen, a large discontinuous control gain is used for a faster reaching time if the sliding variable is far away from the specified sliding surface where the tracking performance is unsatisfactory.

Remark 4. With the decrease of the variable $|\Delta s_i|$, the function $k_{c3i}/[\eta_i + (1 + 1/|s_i| - \eta_i)e^{-\alpha_i|\Delta s_i|}]$ will converge to $k_{c3i}|s_i|/(|s_i| + 1)$. When the sliding variable is driven onto the desired sliding surface $s_i = 0$ by the subsequent designed control law, the discontinuous control gain will decrease to zero. In this sense, with the sliding variable approaching the desired sliding surface, the discontinuous control gain will gradually decrease to zero to suppress control chattering.

Based on the defined sliding surface in Equation (11), its time derivative can be calculated as:

$$\dot{s}_i = \dot{x}^e_{2i} + k_{c2i}x^e_{2i} + k_{c1i}x^e_{1i}. \tag{14}$$

By recalling the definition of x^e_{1i} and x^e_{2i}, substituting Equations (8) and (12) into Equation (14) leads to:

$$h_{1i}f_i(x) + h_{2i}x_{2i} + g_iB_{ui}u - \ddot{x}^d_{1i} + k_{c2i}x^e_{2i} + k_{c1i}x^e_{1i} = -\frac{k_{c3i}}{\eta_i + (1 + 1/|s_i| - \eta_i)e^{-\alpha_i|\Delta s_i|}}\text{sign}(s_i). \tag{15}$$

Therefore, by solving Equation (15), the corresponding control law can be designed as follows:

$$u = B^+_{ui}g_i^{-1}\left(\ddot{x}^d_{1i} - k_{c1i}x^e_{1i} - k_{c2i}x^e_{2i} - h_{1i}f_i(x) - h_{2i}x_{2i} - k^*_{c3i}\text{sign}(s_i)\right), \tag{16}$$

where $B^+_{ui} = B^T_{ui}(B_{ui}B^T_{ui})^{-1}$ and $k^*_{c3i} = k_{c3i}/[\eta_i + (1 + 1/|s_i| - \eta_i)e^{-\alpha_i|\Delta s_i|}]$.

Furthermore, model uncertainties are common in quadrotor UAVs. They might compromise the tracking control performance of quadrotor UAVs or possibly cause instability of the system. With this in mind, an adaptive control law is developed to compensate model uncertainties as:

$$u = B^+_{ui}g_i^{-1}\left(\ddot{x}^d_{1i} - k_{c1i}x^e_{1i} - k_{c2i}x^e_{2i} - \hat{h}_{1i}f_i(x) - \hat{h}_{2i}x_{2i} - k^*_{c3i}\text{sign}(s_i)\right). \tag{17}$$

The corresponding adaptation laws for estimating the uncertain parameters are then designed in the following manner:

$$\begin{aligned} \dot{\hat{h}}_{1i} &= \gamma_{1i}f_i(x)\Delta s_i \\ \dot{\hat{h}}_{2i} &= \gamma_{2i}x_{2i}\Delta s_i, \end{aligned} \tag{18}$$

where γ_{1i} and γ_{2i} are positive parameters that control the adaptation rate.

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Theorem 1. *Consider an integral-chain nonlinear system in Equation (8) with model uncertainties, given the defined sliding surface in Equation (11) and the proposed reaching law in Equation (12), an adaptive feedback control law is designed in Equation (17) that is updated by Equation (18). Therefore, the desired sliding motion will be obtained and system tracking performance can be ensured with the discontinuous control gain chosen as* $k^*_{c3i} = k_{c3i}/[\eta_i + (1 + 1/|s_i| - \eta_i)e^{-\alpha_i|\Delta s_i|}]$.

Proof. Consider the Lyapunov function as:

$$V_1 = \sum_{i=1}^{3} \frac{1}{2}\left[\Delta s_i^2 + \frac{1}{\gamma_{1i}}(\hat{h}_{1i} - h_{1i})^2 + \frac{1}{\gamma_{2i}}(\hat{h}_{2i} - h_{2i})^2\right]. \tag{19}$$

Then, considering the condition that $\Delta s_i \neq 0$, that is, the sliding variable is outside the boundary layer, and recalling the control law in Equation (17) and the adaptation law in Equation (18), the time derivative of V_1 can be calculated as:

$$\begin{aligned}
\dot{V}_1 &= \sum_{i=1}^{3}\left[\Delta s_i \Delta \dot{s}_i + \frac{1}{\gamma_{1i}}(\hat{h}_{1i} - h_{1i})\dot{\hat{h}}_{1i} + \frac{1}{\gamma_{2i}}(\hat{h}_{2i} - h_{2i})\dot{\hat{h}}_{2i}\right] \\
&= \sum_{i=1}^{3}\left[\Delta s_i\left(h_{1i}f_i(x) + h_{2i}x_{2i} - \hat{h}_{1i}f_i(x) - \hat{h}_{2i}x_{2i} - k^*_{c3i}\mathrm{sign}(s_i)\right) + \frac{1}{\gamma_{1i}}(\hat{h}_{1i} - h_{1i})\dot{\hat{h}}_{1i} + \frac{1}{\gamma_{2i}}(\hat{h}_{2i} - h_{2i})\dot{\hat{h}}_{2i}\right] \\
&= \sum_{i=1}^{3}\left[\left(\hat{h}_{1i} - h_{1i}\right)\left(\frac{\dot{\hat{h}}_{1i}}{\gamma_{1i}} - f_i(x)\Delta s_i\right) + \left(\hat{h}_{2i} - h_{2i}\right)\left(\frac{\dot{\hat{h}}_{2i}}{\gamma_{2i}} - x_{2i}\Delta s_i\right) - k^*_{c3i}\mathrm{sign}(s_i)\Delta s_i\right] \\
&= \sum_{i=1}^{3}\left[-k^*_{c3i}|\Delta s_i|\right] < 0.
\end{aligned} \tag{20}$$

As a result, in the presence of certain level of model uncertainties with bounded Δ_1 and Δ_2, the system's stability can be guaranteed by using the developed control law and adaptation law with reasonable adaptation rate. ■

3.2 | Adaptive fault-tolerant SMC

Considering the occurrence of actuator faults in the quadrotor UAV, the diagonal matrix L_c, which measures the fault severity of the actuators, is not an identity matrix. One way to achieve FTC purpose of compensating the negative effect of actuator faults is to use a fault diagnosis unit to estimate the actual value of L_c and incorporate it with the developed control law. In practice, however, there may be a significant time delay in obtaining the essential fault information. Moreover, there may be fault estimation inaccuracies, which would have an impact on the relevant control performance. In this section, an adaptive FTC technique is proposed for the studied quadrotor UAV to tolerate actuator faults without the need for fault information.

Under this condition, Equation (8) can be rewritten as:

$$\begin{cases}
\dot{x}_{1i} = x_{2i} \\
\dot{x}_{2i} = (h_{1i} + \Delta h_{1i})f_i(x) + (h_{2i} + \Delta h_{2i})x_{2i} + g_i(v_i^d + v_i^e) \\
v_i = B_{ui}I_c u - B_{ui}(I_c - L_c)u,
\end{cases} \tag{21}$$

where I_c is an identity matrix with the same dimension as L_c, $v_i = v_i^d + v_i^e$ denotes the actual virtual control signal generated by the individual actuators under the actuator faulty condition, $v_i^d = B_{ui}I_c u$ represents the desired virtual control signal derived from the developed control law without considering the actuator faults, $v_i^e = -B_{ui}(I_c - L_c)u$ represents the virtual control signal error due to the occurrence of actuator faults.

As can be observed from Equation (21), to mitigate the negative effects of virtual control signal error induced by actuator faults while maintaining the desired system tracking control performance, one intuitive way is to adaptively adjust the parameter g_i. In this case, let $g_i v_i^e = \tilde{g}_i v_i^d$, and Equation (21) can be reconstructed as follows:

$$\dot{x}_{2i} = (h_{1i} + \Delta h_{1i})f_i(x) + (h_{2i} + \Delta h_{2i})x_{2i} + (g_i + \tilde{g}_i)v_i^d = (h_{1i} + \Delta h_{1i})f_i(x) + (h_{2i} + \Delta h_{2i})x_{2i} + \hat{g}_i v_i^d, \tag{22}$$

where $\hat{g}_i$ represents the estimation of g_i, and $\tilde{g}_i = \hat{g}_i - g_i$ denotes the estimation error.

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Under this condition, instead of using g_i for developing the feedback control law, the estimated parameter $\hat{g}_i$ should be employed to compensate the actuator faults and maintain the original tracking control performance of the closed-loop system. Moreover, recalling the developed control law in Equation (17), the parameters $\hat{h}_{1i}$ and $\hat{h}_{2i}$ need to be utilized in conjunction with $\hat{g}_i$ to develop the control law for compensation of actuator faults and model uncertainties. Therefore, by denoting $\hat{\Gamma}_i = \hat{g}_i^{-1}\hat{h}_{1i}$, $\hat{\Upsilon}_i = \hat{g}_i^{-1}\hat{h}_{2i}$, and $\hat{\Psi}_i = \hat{g}_i^{-1}$, the feedback control law can be developed as:

$$u = B_{ui}^{+}\hat{\Psi}_i\left(\ddot{x}_{1i}^{d} - k_{c1i}x_{1i}^{e} - k_{c2i}x_{2i}^{e} - k_{c3i}^{*}\mathrm{sign}(s_i)\right) - B_{ui}^{+}\hat{\Gamma}_i f_i(x) - B_{ui}^{+}\hat{\Upsilon}_i x_{2i}. \tag{23}$$

The corresponding adaptation laws for uncertain parameter estimation are designed as:

$$\begin{aligned}
\dot{\hat{\Gamma}}_i &= \rho_{1i} f_i(x)\Delta s_i,\\
\dot{\hat{\Upsilon}}_i &= \rho_{2i} x_{2i}\Delta s_i,\\
\dot{\hat{\Psi}}_i &= \rho_{3i}\left(k_{c1i}x_{1i}^{e} + k_{c2i}x_{2i}^{e} - \ddot{x}_{1i}^{d} + k_{c3i}^{*}\mathrm{sign}(s_i)\right)\Delta s_i,
\end{aligned} \tag{24}$$

where ρ_{1i}, ρ_{2i}, and ρ_{3i} are positive parameters for tuning the adaptation rate.

Remark 5. As can be seen from the developed feedback control law in Equation (23), rather than adaptively adjusting the discontinuous control gain to compensate actuator faults, $\hat{\Psi}_i$ and k_{c3i}^{*} are treaded together as the discontinuous control gain and $\hat{\Psi}_i$ is adaptively estimated. Since $\hat{\Psi}_i$ is employed in both continuous and discontinuous control components, both of them will contribute to compensate the actuator faults after fault occurrence. In this sense, the utilization of the discontinuous control component will be decreased, which can contribute to preventing control chattering.

Theorem 2. *Consider an integral-chain nonlinear quadrotor UAV system in Equation (8) with actuator faults and model uncertainties. Given the proposed reaching law in Equation (12), by utilizing the developed feedback control law in Equation (23) and the adaptation law in Equation (24), the desired system tracking control performance can be achieved in spite of actuator faults and model uncertainties.*

Proof. A Lyapunov function is selected as:

$$V_2 = \sum_{i=1}^{3}\frac{1}{2}\left[\Delta s_i^2 + \frac{1}{\rho_{1i}\Psi_i}\tilde{\Gamma}_i^2 + \frac{1}{\rho_{2i}\Psi_i}\tilde{\Upsilon}_i^2 + \frac{1}{\rho_{3i}\Psi_i}\tilde{\Psi}_i^2\right], \tag{25}$$

where $\tilde{\Gamma}_i = \hat{\Gamma}_i - \Gamma_i$, $\tilde{\Upsilon}_i = \hat{\Upsilon}_i - \Upsilon_i$, and $\tilde{\Psi}_i = \hat{\Psi}_i - \Psi_i$.

For simplicity of analysis, V_2 is divided into four components as:

$$V_2 = V_{21} + V_{22} + V_{23} + V_{24}, \tag{26}$$

where $V_{21} = \sum_{i=1}^{3}\Delta s_i^2/2$, $V_{22} = \sum_{i=1}^{3}\tilde{\Gamma}_i^2/(2\rho_{1i}\Psi_i)$, $V_{23} = \sum_{i=1}^{3}\tilde{\Upsilon}_i^2/(2\rho_{2i}\Psi_i)$, and $V_{24} = \sum_{i=1}^{3}\tilde{\Psi}_i^2/(2\rho_{3i}\Psi_i)$.

First, recalling Equation (23), the time derivative of V_{21} is calculated as:

$$\begin{aligned}
\dot{V}_{21} &= \sum_{i=1}^{3}\Delta s_i[\dot{x}_{2i}^{e} + k_{c2i}x_{2i}^{e} + k_{c1i}x_{1i}^{e}]\\
&= \sum_{i=1}^{3}\Delta s_i[\Psi_i^{-1}\Gamma_i f_i(x) + \Psi_i^{-1}\Upsilon_i x_{2i} + \Psi_i^{-1}[\hat{\Psi}_i(\ddot{x}_{1i}^{d} - k_{c1i}x_{1i}^{e} - k_{c2i}x_{2i}^{e} - k_{c3i}^{*}\mathrm{sign}(s_i))\\
&\quad - \hat{\Gamma}_i f_i(x) - \hat{\Upsilon}_i x_{2i}] - \ddot{x}_{1i}^{d} + k_{c2i}x_{2i}^{e} + k_{c1i}x_{1i}^{e}]\\
&= \sum_{i=1}^{3}\Delta s_i[\Psi_i^{-1}f_i(x)(\Gamma_i - \hat{\Gamma}_i) + \Psi_i^{-1}x_{2i}(\Upsilon_i - \hat{\Upsilon}_i) - k_{c3i}^{*}\mathrm{sign}(s_i)\\
&\quad + (1 - \Psi_i^{-1}\hat{\Psi}_i)(k_{c2i}x_{2i}^{e} + k_{c1i}x_{1i}^{e} - \ddot{x}_{1i}^{d} + k_{c3i}^{*}\mathrm{sign}(s_i))].
\end{aligned} \tag{27}$$

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Then, employing the adaptation law in Equation (24), one can obtain that:

$$\begin{aligned}\dot{V}_{22}+\dot{V}_{23}+\dot{V}_{24} &= \sum_{i=1}^{3}[\Psi_i^{-1}(\hat{\Gamma}_i-\Gamma_i)\dot{\hat{\Gamma}}_i/\rho_{1i}+\Psi_i^{-1}(\hat{\Upsilon}_i-\Upsilon_i)\dot{\hat{\Upsilon}}_i/\rho_{2i}+\Psi_i^{-1}(\hat{\Psi}_i-\Psi_i)\dot{\hat{\Psi}}_i/\rho_{3i}]\\ &= \sum_{i=1}^{3}[\Psi_i^{-1}(\hat{\Psi}_i-\Psi_i)(k_{c1i}x_{1i}^e+k_{c2i}x_{2i}^e-\ddot{x}_{1i}^d+k_{c3i}^*\mathrm{sign}(s_i))\Delta s_i\\ &\quad+\Psi_i^{-1}(\hat{\Gamma}_i-\Gamma_i)f_i(x)\Delta s_i+\Psi_i^{-1}(\hat{\Upsilon}_i-\Upsilon_i)x_{2i}\Delta s_i].\end{aligned} \tag{28}$$

Combining Equations (27) and (28) and considering the condition that $\Delta s_i \neq 0$, it gives:

$$\begin{aligned}\dot{V}_2 &= \sum_{i=1}^{3}\Delta s_i[\Psi_i^{-1}(\Gamma_i-\hat{\Gamma}_i)f_i(x)+\Psi_i^{-1}(\Upsilon_i-\hat{\Upsilon}_i)x_{2i}+(1-\Psi_i^{-1}\hat{\Psi}_i)(k_{c2i}x_{2i}^e+k_{c1i}x_{1i}^e-\ddot{x}_{1i}^d+k_{c3i}^*\mathrm{sign}(s_i))-k_{c3i}^*\mathrm{sign}(s_i)\\ &\quad+\Psi_i^{-1}(\hat{\Psi}_i-\Psi_i)(k_{c1i}x_{1i}^e+k_{c2i}x_{2i}^e-\ddot{x}_{1i}^d+k_{c3i}^*\mathrm{sign}(s_i))+\Psi_i^{-1}(\hat{\Gamma}_i-\Gamma_i)f_i(x)+\Psi_i^{-1}(\hat{\Upsilon}_i-\Upsilon_i)x_{2i}]\\ &= \sum_{i=1}^{3}\Delta s_i[\Psi_i^{-1}(\Gamma_i-\hat{\Gamma}_i)f_i(x)+\Psi_i^{-1}(\Upsilon_i-\hat{\Upsilon}_i)x_{2i}+(1-\Psi_i^{-1}\hat{\Psi}_i)(k_{c2i}x_{2i}^e+k_{c1i}x_{1i}^e-\ddot{x}_{1i}^d+k_{c3i}^*\mathrm{sign}(s_i))-k_{c3i}^*\mathrm{sign}(s_i)\\ &\quad-(1-\Psi_i^{-1}\hat{\Psi}_i)(k_{c2i}x_{2i}^e+k_{c1i}x_{1i}^e-\ddot{x}_{1i}^d+k_{c3i}^*\mathrm{sign}(s_i))-\Psi_i^{-1}(\Gamma_i-\hat{\Gamma}_i)f_i(x)-\Psi_i^{-1}(\Upsilon_i-\hat{\Upsilon}_i)x_{2i}]\\ &= \sum_{i=1}^{3}\Delta s_i[-k_{c3i}^*\mathrm{sign}(s_i)].\end{aligned} \tag{29}$$

For the above equation, we have the following two conditions. If $s_i > \Phi_i$, then $\mathrm{sign}(s_i) = 1$, $\mathrm{sat}(s_i) = 1$, and $\mathrm{sign}(s_i)\Delta s_i = \Delta s_i = s_i - \Phi_i\mathrm{sat}(s_i) = s_i - \Phi_i > 0$. In this case, we can get that $\mathrm{sign}(s_i)\Delta s_i = \Delta s_i = |\Delta s_i|$. If $s_i < -\Phi_i$, then $\mathrm{sign}(s_i) = -1$, $\mathrm{sat}(s_i) = -1$, and $\mathrm{sign}(s_i)\Delta s_i = -\Delta s_i = -(s_i - \Phi_i\mathrm{sat}(s_i)) = -(s_i + \Phi_i) > 0$. In this case, we can obtain that $\mathrm{sign}(s_i)\Delta s_i = -\Delta s_i = |\Delta s_i|$. Based on above analysis, we can get that no matter Δs_i is positive or negative, we can always guarantee that $\Delta s_i\mathrm{sign}(s_i) = |\Delta s_i|$. Therefore, Equation (29) can be rewritten as:

$$\dot{V}_2 = \sum_{i=1}^{3}\left[-k_{c3i}^*|\Delta s_i|\right] < 0. \tag{30}$$

As a result, under the influence of certain level of actuator faults and model uncertainties with bounded Δ_1 and Δ_2, system stability can be maintained from any initial conditions by using the proposed adaptive control strategy with reasonable adaptation rate. ■

Remark 6. As per the definition of the designed variable Δs_i, by employing this variable for constructing the adaptation law in Equation (24), the adaptation will be ceased as long as the sliding variable is inside the defined boundary, which can help to avoid overestimation of the discontinuous control gain. As a result, the proposed adaptive scheme can avoid control chattering as compared to the existing adaptive SMC methods in the literature, in which it is difficult to guarantee the adaptation convergence due to the use of sliding variable for adaptation law formulation and the inevitable tracking errors.

4 | SIMULATION RESULTS AND DISCUSSIONS

A series of simulation tests are undertaken to quantitatively examine the effectiveness of the proposed adaptive fault-tolerant control method (PAFTC) for mitigating actuator faults and model uncertainties of the quadrotor UAV. Table 1 lists the physical specifications of the quadrotor UAV under investigation. For the sake of comparison and demonstrating the advantages of PAFTC, the performance of the SMC constructed with the proposed reaching law (SMCRL) and a nominal adaptive SMC whose adaptation law is formulated with sliding variable (NASMC)[42] are investigated as well. The control parameters of PAFTC are set as follows: $\eta_1 = 0.1, \alpha_1 = 10, k_{c11} = 100, k_{c21} = 20, k_{c31} = 1, \Phi_1 = 0.1, \rho_{11} = \rho_{21} = \rho_{31} = 1, \hat{\Gamma}_1(0) = \hat{\Upsilon}_1(0) = 0, \hat{\Psi}_1(0) = 0.03, \eta_2 = 0.1, \alpha_2 = 10, k_{c12} = 100, k_{c22} = 20, k_{c32} = 1, \Phi_2 = 0.1, \rho_{12} = \rho_{22} = \rho_{32} = 1, \hat{\Gamma}_2(0) = \hat{\Upsilon}_2(0) = 0, \hat{\Psi}_2(0) = 0.03, \eta_3 = 0.1, \alpha_3 = 10, k_{c13} = 100, k_{c23} = 20, k_{c33} = 1,$

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TABLE 1 Physical specifications of the researched quadrotor UAV.

Symbols	Interpretations	Values
$I = \mathrm{diag}([I_{xx}, I_{yy}, I_{zz}])$	Moment of inertia	$I = \mathrm{diag}([0.03, 0.03, 0.04])$ kg · m^2
L	Distance between motor and center of gravity	0.2 m
K_u	Thrust gain	120 N
K_y	Torque gain	4 N · m

$\Phi_3 = 0.1, \rho_{13} = \rho_{23} = \rho_{33} = 1, \hat{\Gamma}_3(0) = \hat{\Upsilon}_3(0) = 0, \hat{\Psi}_3(0) = 0.04$. The non-adaptive parameters of SMCRL and NASMC are chosen the same as those of PAFTC. The following two scenarios are used to demonstrate the effectiveness of the proposed control strategy. Furthermore, for the purpose of quantitatively evaluating the tracking performance of the demonstrated control approaches, an index indicating the root mean squared error (RMSE) of the attitude tracking is specified as:

$$e_{\mathrm{RMSE}} = \sqrt{\frac{1}{t_1 - t_0}\int_{t_0}^{t_1} |x_{1i}^d - x_{1i}|^2 d\tau}, \tag{31}$$

where $[t_0, t_1]$ spans the entire time frame of the simulation test.

In Scenario 1, a 20% loss of control effectiveness fault is injected into actuator #1 at 10 s and actuator #4 at 20 s, the uncertain parameters for drag forces are set to $k_{di} = 0.2$ $(i = 1, 2, 3)$, and the model uncertainties for inertial moments are set to $\Delta I = 0.5I$ and injected into the attitude system at 15 s. The detailed tracking performance of roll, pitch, and yaw motions are demonstrated in Figure 3. After faults occurrence, the PAFTC can instantaneously change both the discontinuous control gain as shown in Figure 3D and the adaptive control parameters as illustrated in Figure 4 to mitigate the negative impact caused by actuator faults and retain the original system tracking performance. In addition, both the compared SMCRL and NASMC can retain the appropriate system tracking performance as well. However, as shown in Figure 5, the compared NASMC induces control chattering because of the employment of a sliding variable in the adaptation law formulation. Figure 3 shows that the PAFTC developed in this article outperforms the compared SMCRL and NASMC with less tracking errors and avoidance of control chattering. Moreover, both the PAFTC and SMCRL use the same reaching law for SMC development, however, the PAFTC has less tracking errors than the SMCRL. This is because in the PAFTC, the developed adaptation law is incorporated in both continuous and discontinuous control components of SMC, which can contribute to compensating actuator faults without merely relying on the robustness of the discontinuous control component of SMC.

In Scenario 2, a larger actuator fault with 35% loss of control effectiveness is injected into actuator #1 at 10 s and actuator #4 at 20 s, respectively, to further test the effectiveness and capability of the proposed control method. Moreover, when compared to Scenario 1, the uncertain parameters for drag forces are adjusted with a bigger amplitude and time-varying property. The uncertain parameters are set as $k_{di} = 10\sin(0.2\pi t)$ $(i = 1, 2, 3)$, and the uncertain inertial moments are the same as those in Scenario 1. The corresponding tracking performance of roll, pitch, and yaw motions are demonstrated in Figure 6. Compared to Scenario 1, due to the increased severity of actuator faults and model uncertainties, the compared SMCRL is more affected and exhibits large tracking errors. Nevertheless, under this faulty and uncertain condition, although the compared NASMC can track the desired trajectory, it exhibits oscillatory tracking errors. As can be observed from Figure 8, the compared NASMC uses more control effort and induces severe control chattering. In contrast, the PAFTC is still able to retain the necessary tracking performance with modest tracking deviations. Figures 7 and 6D demonstrate the change of adaptive control parameters and discontinuous control gain, which are bigger than those in Scenario 1 because of the increased severity of actuator faults and model uncertainties.

Table 2 summarizes the performance indices related to the roll, pitch, and yaw tracking. Taking roll motion control in Scenario 1 as an example, compared to SMCRL and NASMC, the PAFTC has improved the RMSE of roll angle from 0.1466° and 0.1297° to 0.0611° with the enhancement rate of 58.32% and 52.89%, respectively. As per the performance indices in Table 2, the PAFTC can ensure the desired tracking performance of the quadrotor UAV even under the unfavorable conditions involving actuator faults, inertial moment variations, and time-varying drag forces. In addition, the quantitative comparison results confirm the effectiveness and superiority of the PAFTC.

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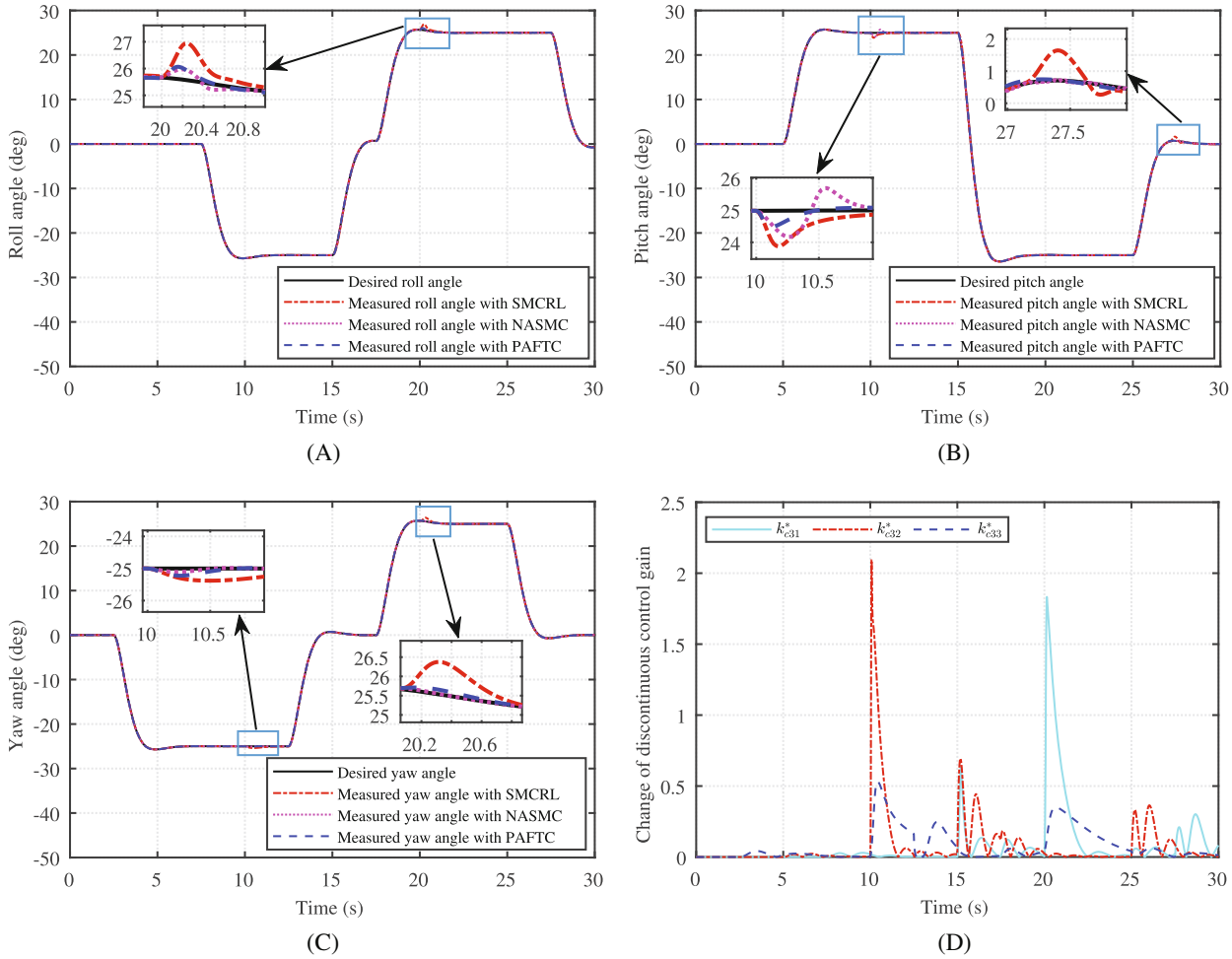

FIGURE 3 Tracking performance of quadrotor UAV in Scenario 1. (A) Roll motion responses; (B) pitch motion responses; (C) yaw motion responses; (D) change of discontinuous control gain.

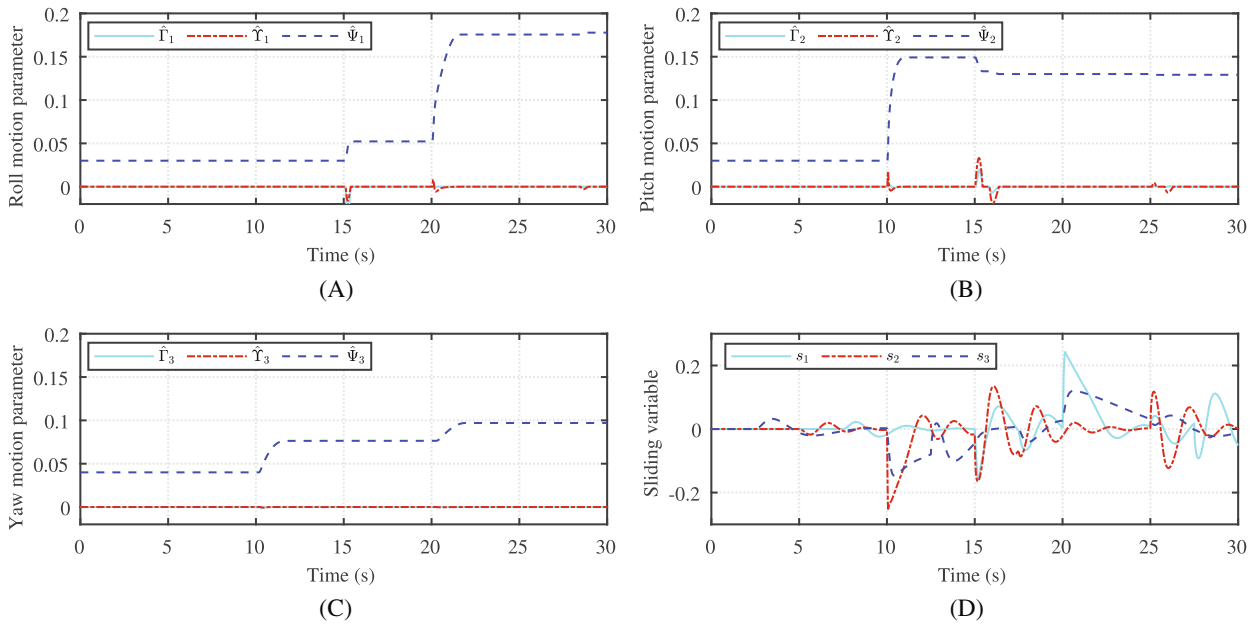

FIGURE 4 Change of adaptive control parameter and sliding variable of PAFTC in Scenario 1. (A) Roll motion related adaptive control parameter; (B) pitch motion related adaptive control parameter; (C) yaw motion related adaptive control parameter; (D) change of sliding variable.

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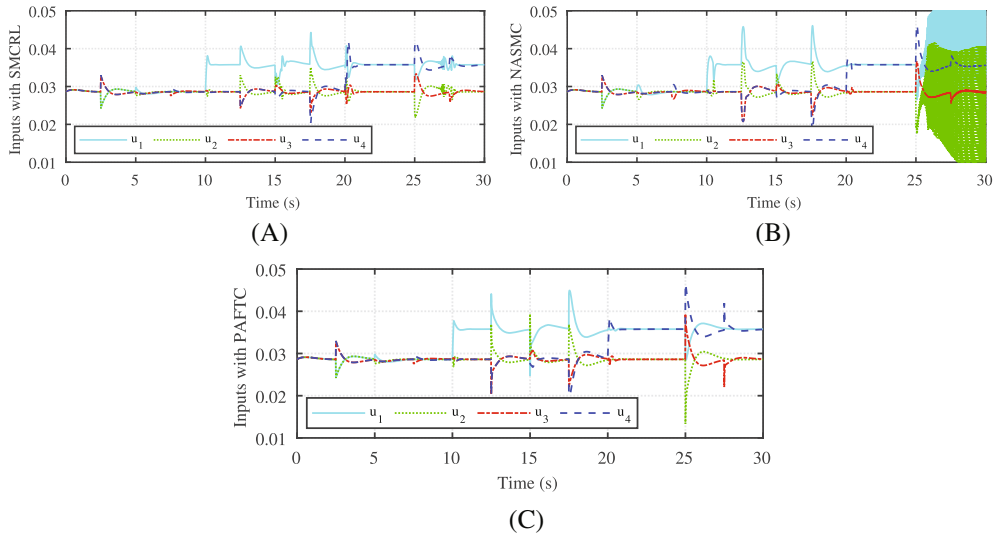

FIGURE 5 Control inputs of quadrotor UAV in Scenario 1. (A) control inputs with SMCRL; (B) control inputs with NASMC; (C) control inputs with PAFTC.

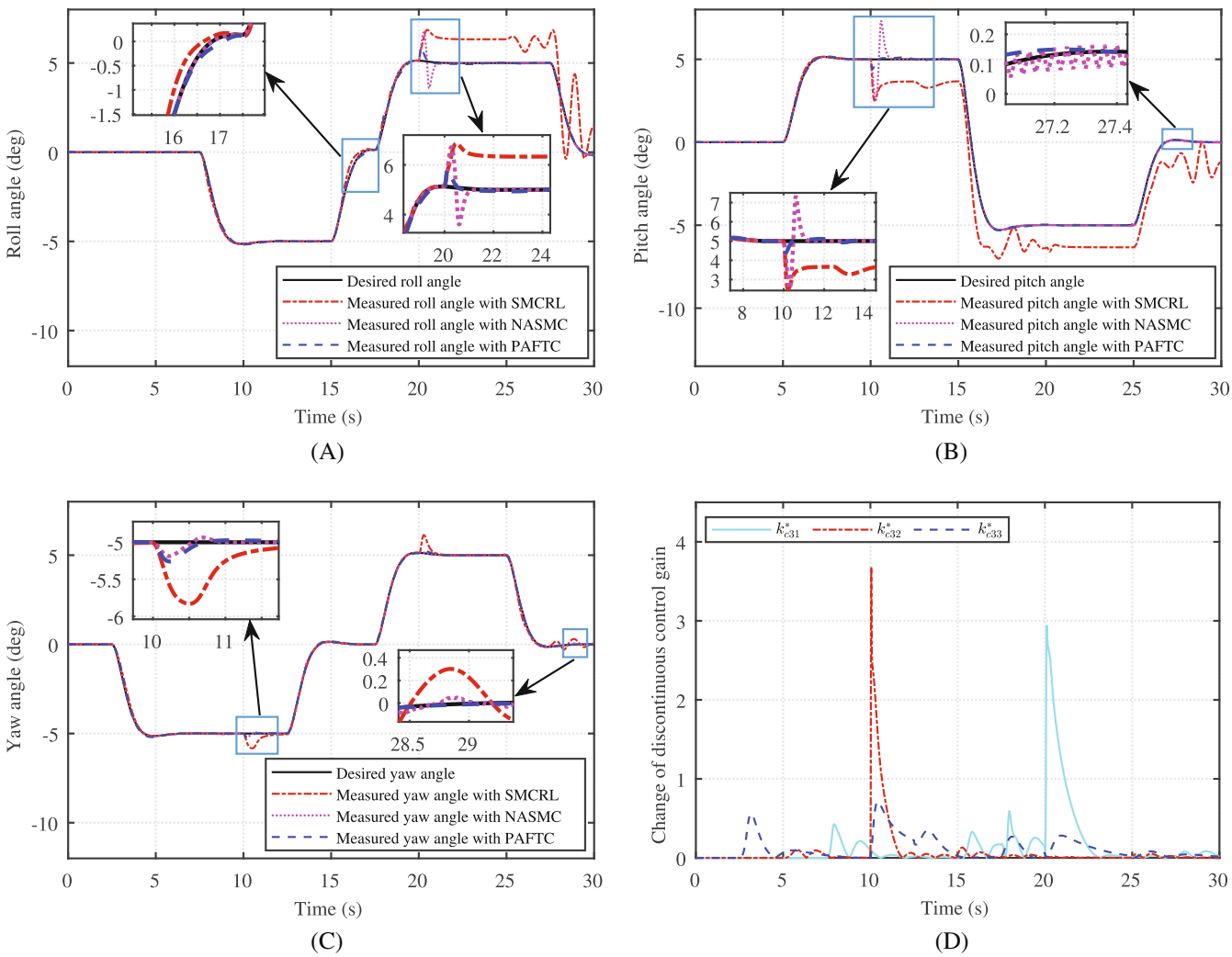

FIGURE 6 Tracking performance of quadrotor UAV in Scenario 2. (A) Roll motion responses; (B) pitch motion responses; (C) yaw motion responses; (D) change of discontinuous control gain.

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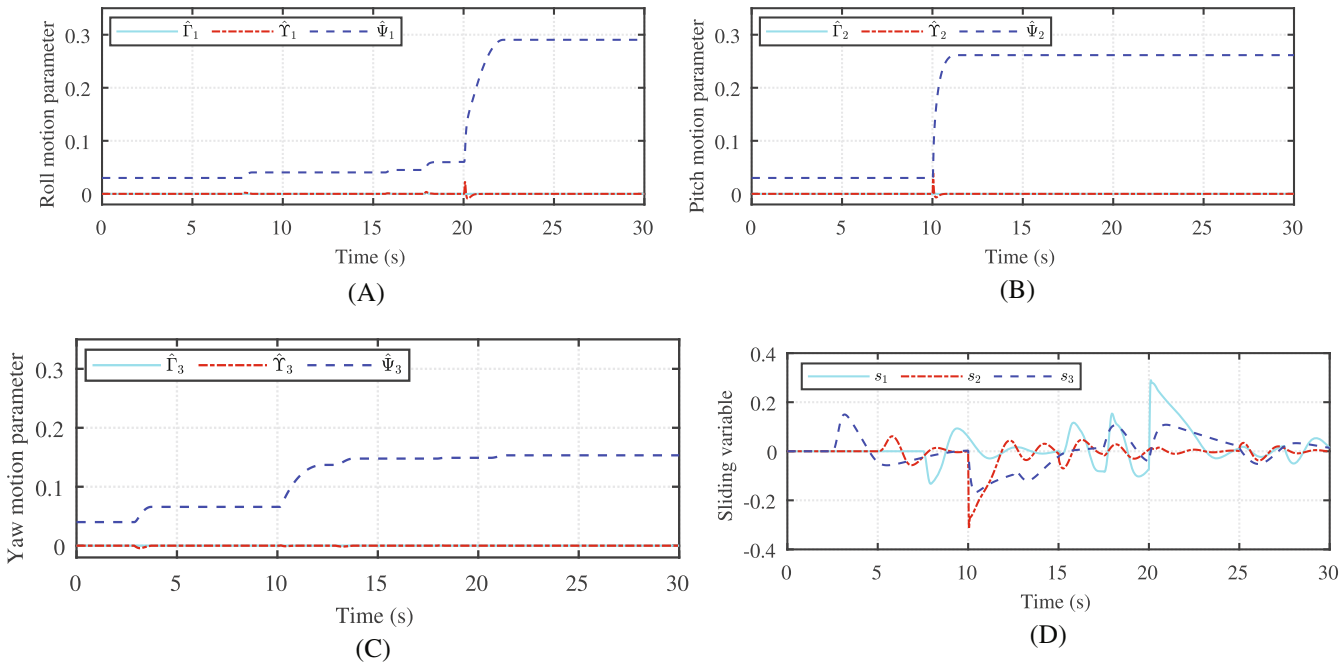

FIGURE 7 Change of adaptive control parameter and sliding variable of PAFTC in Scenario 2. (A) Roll motion related adaptive control parameter; (B) pitch motion related adaptive control parameter; (C) yaw motion related adaptive control parameter; (D) change of sliding variable.

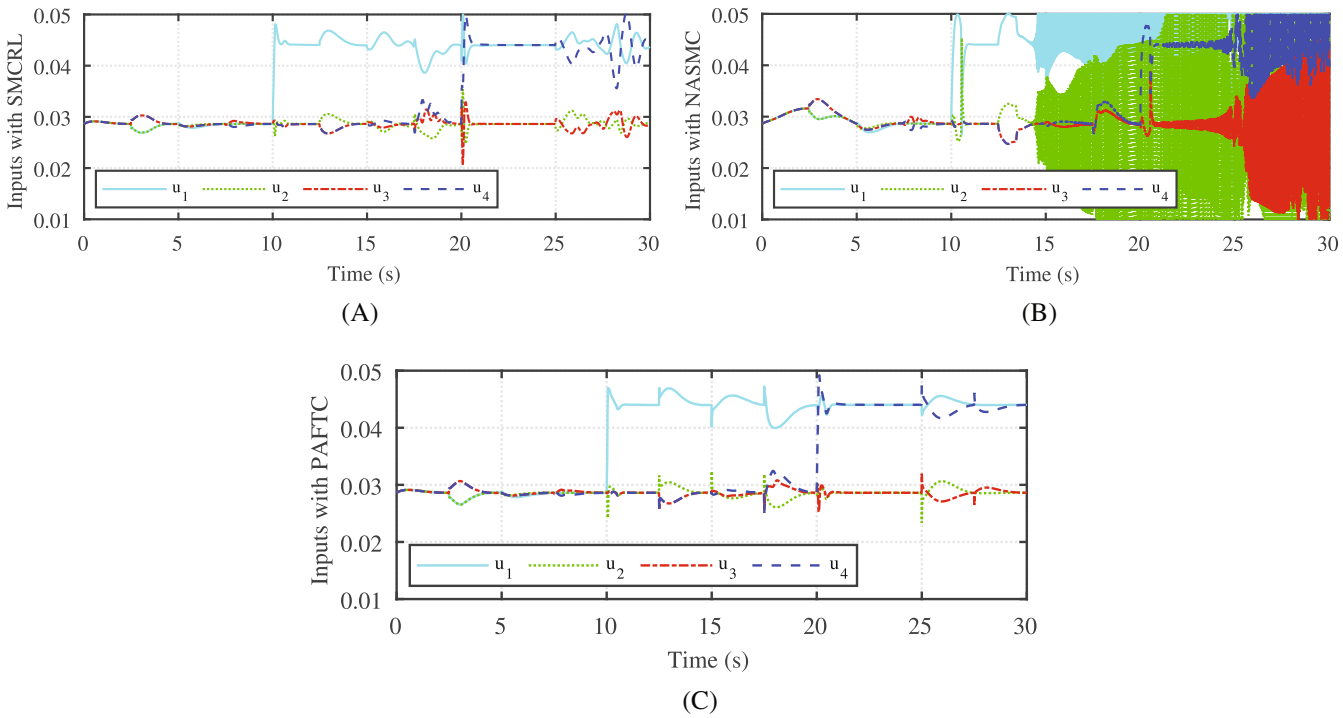

FIGURE 8 Control inputs of quadrotor UAV in Scenario 2. (A) Control inputs with SMCRL; (B) control inputs with NASMC; (C) control inputs with PAFTC.

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TABLE 2 Comparison of performance indices.

Scenario		SMCRL	NASMC	PAFTC	Enhancement rate
Scenario 1	RMSE of ϕ	0.1466°	0.1297°	0.0611°	58.32%&52.89%
	RMSE of θ	0.1558°	0.1169°	0.0603°	61.29%&48.42%
	RMSE of ψ	0.1110°	0.0372°	0.0349°	68.56%&6.18%
Scenario 2	RMSE of ϕ	0.8525°	0.1784°	0.0834°	90.22%&53.25%
	RMSE of θ	1.1838°	0.2750°	0.0669°	94.35%&75.67%
	RMSE of ψ	0.1707°	0.0443°	0.0379°	77.79%&14.45%

5 | CONCLUSIONS

An adaptive sliding mode FTC strategy for a quadrotor UAV is proposed in this article to compensate both actuator faults and model uncertainties. First, a new reaching law is proposed to achieve a fast reaching time while avoiding control chattering. Based on the proposed reaching law, a baseline SMC law is developed, which can effectively suppress the unexpected control chattering. Then, to further address the problem of compensating actuator faults and model uncertainties, an adaptive FTC scheme is proposed to mitigate the adverse impact of both actuator faults and model uncertainties. With the aid of the proposed adaptation law, overestimation of the uncertain adaptive control parameters can be prevented. Finally, the effectiveness and advantages of the proposed control strategy are demonstrated in a series of comparative simulation tests, in which the quadrotor UAV is exposed to different level of actuator faults, inertial moment variations, and time-varying drag forces.

CONFLICT OF INTEREST STATEMENT

The authors declare that there is no potential conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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